. The first excited state's mean lifetime is . Using ,
how long must the magnetic field region be to bring the atom to rest?

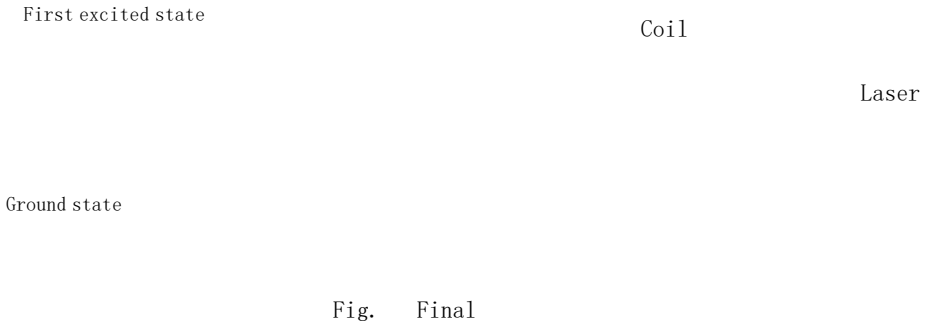


Fig. Final